

Vatsa Joshi

📍 Bengaluru, Karnataka 📞 +91 9737557799 ✉️ vatsajoshi2@gmail.com
🌐 [linkedin.com/in/vatsa-joshi](https://www.linkedin.com/in/vatsa-joshi) 🏠 github.com/Vatsa10 🌐 vatsa.online

Technical Skills

Languages	Python, JavaScript/TypeScript
AI/ML	PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers, Unsloth
LLM & Agents	LangChain, LangGraph, CrewAI, RAG, CAG, LoRA/QLoRA, FAISS, Groq
Vision	OpenCV, YOLO, SSD, PaddleOCR, Diffusers
Backend / Cloud	FastAPI, NestJS, Node.js, Docker, AWS, PostgreSQL, MongoDB, Elasticsearch, GraphDB
Frontend	Next.js, Astro

Work Experience

Software Engineer Trainee (AI)

Jan 2026 – May 2026

Promact · Vadodara

- Contributed to **BharatLaw.ai**, an AI-powered legal technology product for Indian law firms.
- Reduced legal NER extraction cost by **25×** on an 80-lakh judgment corpus by replacing a 7-stage LLM pipeline with a 2-call Gemini-2.5-Flash-Lite architecture.
- Achieved **98% document-level zero-failure** across 40-lakh test rows using DB/CSV-grounded structural extraction and windowed precedent extraction.
- Built AI-driven document management workflows for translation, parsing, indexing, and retrieval.
- Developed multi-agent legal systems for petition drafting, judge profiling, litigation strategy, lawyer recommendations, and compatibility analysis.

Full Stack Developer — Internship

Jun 2025 – Dec 2025

Hiraeth Tech · Vadodara

- Built scalable products using NestJS, Next.js, PostgreSQL, API integration, and version-controlled deployments.

Deep Learning Engineer — Internship

Mar 2025 – May 2025

Prodigal AI · Remote

- Trained CNN image classifiers achieving **92% accuracy**; fine-tuned transformer models for NLP tasks.
- Developed real-time object detection pipelines with YOLO and SSD; contributed to LLM architecture for a finance firm.

AI/ML Engineer — Internship

Feb 2025 – Apr 2025

Sparks To Ideas · Remote

- Trained an XGBoost customer churn model achieving **95% accuracy**.
- Built an AI content intelligence pipeline with LLM parsing and FAISS HNSW multilingual deduplication.

Selected Projects

Booky | Multi-Agent, LLM, Email

Conversational appointment-booking system with availability checks and automated confirmations.

FloWork | LangGraph, Groq, Next.js, Flask

N8N-inspired agentic workflow automation platform with visual node builder and AI conditional routing.

Hiring-Assistant | CrewAI, Gemini, Streamlit

Recruitment chatbot for screening, dynamic technical assessment, and sentiment-based evaluation.

PersonAI | Groq, LangChain, FastMCP, Next.js

Agent automating Gmail, Calendar, and Tasks through Workspace APIs with OAuth 2.1.

LegalOCR | PaddleOCR, FastAPI, Elasticsearch

Legal document parser with 22+ language OCR, semantic chunking, and Elasticsearch indexing.

Synapse | Next.js, Qdrant, Redis

Multi-channel AI memory unifying web, WhatsApp, email, and phone conversations.

Hindi2Guj-Qwen3.5 | Qwen, DeepSpeed, Accelerate

Fine-tuned Qwen3.5-0.8B for Hindi-to-Gujarati translation using ZeRO distributed training.

Gemma 3 Fine-tune | Unsloth, LoRA, HuggingFace

Fine-tuned Gemma 3 4B on FineTome-100k for instruction-following.

Rustformer | Rust, Transformers

Transformer architecture implemented from scratch in Rust for inference research.

Deepfake Video Detection | Deep Learning, Vision

2nd place at Tinkerthon 3.0 plus 1-year SSIP Gujarat grant.

AarogyaKosh | React Native, FastAPI, Docker

Mobile health app translating lab reports into plain language using on-device medical transformers.

TradeXpert | Next.js, Agentic AI, Inngest

Agentic financial intelligence platform with market data and background analysis pipelines.

Additional Projects

Ecommerce Agent | LangGraph, LLM, WhatsApp — Multi-agent WhatsApp shopping assistant for discovery, comparison, and purchase decisions.

MultiModal-RAG | RAG, Multimodal, LLM — Cross-modal retrieval pipeline supporting text, image, and document inputs.

Email Voice AI Agent | LangGraph, Deepgram, Gmail API — Voice-first inbox assistant for reading, summarizing, starring, and composing email.

Multi-Agent Finance Advisor | RAG, Multi-Agent, LLM — Converts financial filings and earnings transcripts into actionable intelligence.

Flash-Banana | Next.js, Firebase, Genkit — Multimodal creative automation suite for image editing, ads, and social media content.

Education

B.Tech in Computer Science

Navrachana University, Vadodara

Aug 2022 – Sep 2026

Awards

- **2nd Place** — **Tinkerthon 3.0 Hackathon** (Feb 2025) · Deepfake Video Detection System.
- **1-Year Research Grant** — **SSIP Gujarat** · Awarded to further develop Deepfake Detection.
- **Top 5 out of 3300 teams** in Odoo x NMIT Hackathon.

Publications

- **EfficientNetB7 & Bi-LSTM for Deepfake Detection** — IEEE Xplore. DOI: 10.1109/CCIC68129.2026.11486188
- **Report & Analysis of Rabin-Karp Algorithm** — ResearchGate. DOI: 10.13140/RG.2.2.24592.01283
- **Blockchain Technology** — ResearchGate. DOI: 10.13140/RG.2.2.32465.75365